

CHIRADZULU DISTRICT MOCK**2023 MALAWI SCHOOL CERTIFICATE MOCK EXAMINATIONS****CHEMISTRY****Subject Number: M038/I****Monday, 27 March, 2023.****Time Allowed: 2 hours****PAPER I****2:30 – 4:30 pm****(100 marks)****Instructions**

1. This paper contains 12 printed pages. Please check.
2. Write your name at the top of each page.
3. This paper contains **two** sections **A** and **B**.
4. Answer **all** the **thirteen** questions in the spaces provided on each question.
5. Use of electronic calculators is allowed.
6. The maximum number of marks for each answer is indicated against each question.
7. In the table provided on this page, **tick** against the number of the question you have answered.

Question Number	Tick if answered	Do not write in these columns	
1.			
2.			
3.			
4.			
5.			
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7.			
8.			
9.			
10.			
11.			
12.			
13.			

Turn over

Section A (70 marks)

Answer all the questions

1. a. i. State **two** cations that causes water hardness.

(2 marks)

- ii. Explain how ion – exchange is used to remove water hardness.

(3 marks)

- iii. Give any **one** major benefit of treating waste before disposal.

(1 mark)

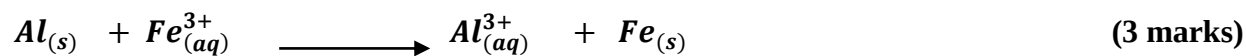
2. a. i. Give any **two** ways of expressing concentration of a solution.

(2 marks)

- b. i. State any **one** factors that affect preferential discharge in an electrolysis.

(1 marks)

- b. Balance the following reaction



3. a. Calculate the oxidation number of sulphur (S) in H_2SO_4

(2 marks)

Figure 1 is a bar which supports a bridge. The iron bar is connected to a piece of magnesium metal

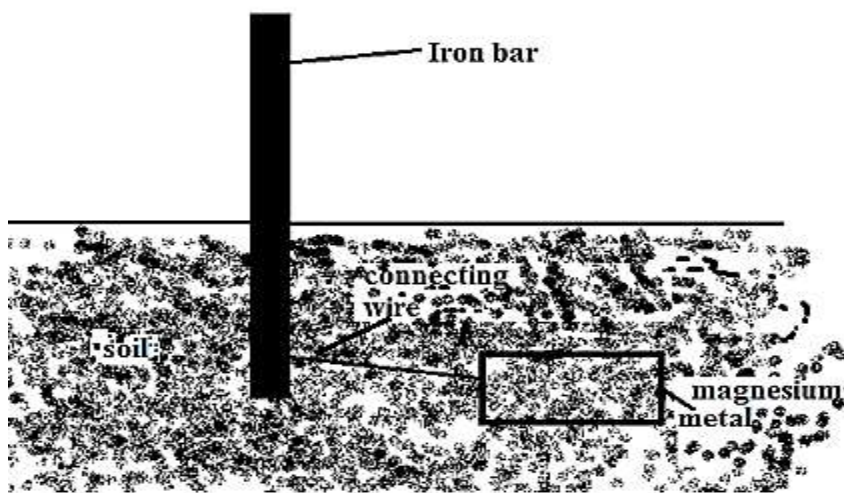


Figure 1

- b. Explain why it is necessary to connect the piece of magnesium to the iron bar

(3 marks)

4. a. Hydrogen gas (H_2) and Chlorine (Cl_2) gas react to form Hydrogen chloride (HCl).



If the total energy required to break the reactants bonds is 678kJ and total energy produced is 862kJ,

- i. is the reaction endothermic or exothermic?

(1 mark)

- ii. Draw the energy level diagram for the reaction.

(3 marks)

- b. Find the molecular formula of a compound of mass 78g and the following percentage composition by mass: C =92.3% and H=7.7%. (RAM: C =12, H =1)

(4 marks)

5. a. What are biodegradable wastes?

(1 mark)

- b. Describe how the presence of Sulphur dioxide (SO_2) can be tested using Potassium dichromate (VI) paper.

(3 marks)

6. a. i . Mention **any two** similarities between silicon oxide and diamond.

(2 marks)

ii. Write a cross and dot diagram for ammonium ion (NH_4^+).

(3 marks)

b. Explain why covalent compounds are more volatile than ionic compounds

(3 marks)

c. Explain how nitrogen gas is used in food packaging.

(3 marks)

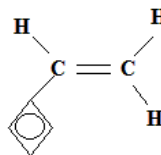
7. a. i. What are position isomers?

(1 mark)

ii. Draw **all** the position isomer butane (C_4H_{10})

(3 marks)

b. Describe how polystyrene is formed from phenylethene



(6 marks)

8. a. State the natural source of the following alkanoic acids

i. Methanoic acid _____

ii. Butanoic acid _____

(2 marks)

b. Draw a tertiary alkanol with 4 carbon atoms

(2 marks)

9. a. State any **two** differences between thermoplastics and thermosets.

(2 marks)

b. Explain how Tollen's reagent can be used to identify an alkanal.

(3 marks)

c. Butanoic acid (C_3H_7COOH) reacts with ethanol (C_2H_5OH) in the presence of concentrated sulphuric acid and warmed.

i. What name is given to the reaction?

(1 marks)

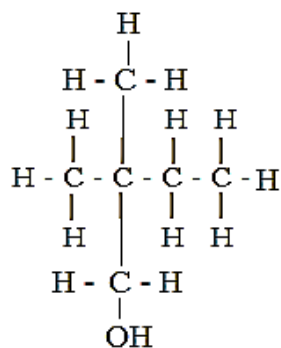
ii. Write the chemical equation of the reaction.

(2 marks)

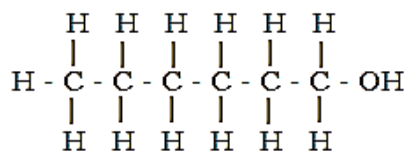
10. a. State any **two** uses of alkanoates.

(2 marks)

b. **Figure 2** are isomers of an alkanol



Isomer A



Isomer B

Figure 2

i. Name the isomer A.

(1 mark)

- ii. Explain why the isomer A has high density than isomer

(2 marks)

- c. **Figure 3** is the skeletal formula of an organic compound.



- i. Name the compound.

_____ (1 mark)

- ii. Write the molecular formula of this compound.

_____ (1 mark)

- iii. In which homogenous series is the compound?

_____ (1 mark)

Section B (30 marks)

Answer all the questions

11. Describe the functions of the three pipes during the extraction of Sulphur.

(6 marks)

b. Describe how electroplating is used in metal extraction

(4 marks)

Name_____School_____

Page 10 of 11

12. a. State any **two** sources of chloroflorocarbons (CFCs).

(2 marks)

b. With an aid of equations, describe how chloroflorocarbons (CFCs) depletes the ozone layer.

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(8 marks)

Page 11 of 11

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END OF QUESTION PAPER

NB: This paper contains 11 printed pages.